

FUNCTIONS OF CEREBELLUM

Functional Division → Main Function

Region	Main Function
Vestibulocerebellum	Balance + eye movements
Spinocerebellum	Ongoing movement + posture + muscle tone
Neocerebellum	Planning + programming + skilled movement + motor learning

1. VESTIBULOCEREBELLUM

A. Balance & Equilibrium

- Maintains **postural balance and equilibrium**.
- Has extensive reciprocal connections with **vestibular nuclei**.
- Receives vestibular inputs from the **inner ear** via vestibular nuclei.

B. Vestibulo-Ocular Reflex (VOR)

- Helps in **learning-induced modification of VOR**.
- Maintains appropriate eye position during head movement.

C. Vestibular Functions

- Balance during movement.
- Vestibular postural reflexes.
- Adjustment of posture and movement with:
 - Head movement
 - Acceleration

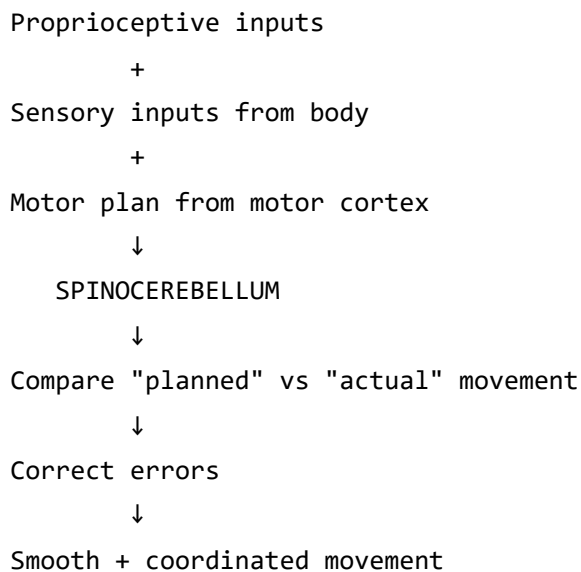
D. Eye Movements

- **Paraflocculus and pyramis** are involved in eyeball movements.
- Particularly involved in **upward eye movement**.
- Contributes to **visual judgment of distance**.

2. SPINOCEREBELLUM

A. Smoothing & Coordination of Movement

Inputs



Therefore

- Coordinates **ongoing movements**.
- Controls:
 - Rate
 - Range
 - Force
 - Direction
 - Termination of movement.

B. Vermal Spinocerebellum → Posture

VERMIS



Brainstem nuclei



Axial + proximal limb muscles



POSTURE

- Controls **posture**.
- Particularly influences **axial and proximal muscles**.

C. Paravermal Spinocerebellum → Skilled Movements

PARAVERMIS



Brainstem motor nuclei



Distal limb muscles



Skilled voluntary movements

- Controls **skilled voluntary movements** of limbs.

D. Muscle Tone

- Cerebellum influences **descending medial motor pathways** through the **fastigial nucleus**.
- Mainly:
 - Vestibulospinal tract → **α motor neurons**
 - Reticulospinal tract → **γ motor neurons**
- Thus, cerebellum contributes to **α-γ co-linkage**.

Clinical

Cerebellar lesion



↓ Excitatory output to brainstem motor nuclei



↓ Muscle tone



HYPOTONIA

- **Stretch reflexes are usually normal.**
- Occasionally → **pendular knee jerk.**

3. NEOCEREBELLUM

A. Planning & Programming of Movement

- Interacts extensively with the **cerebral cortex**.
- Responsible for:
 - Planning of movements
 - Programming of movements
 - Planning of **sequential movements**
 - Timing of sequential movements

Neocerebellum



Motor / Premotor cortex



Planning + Programming



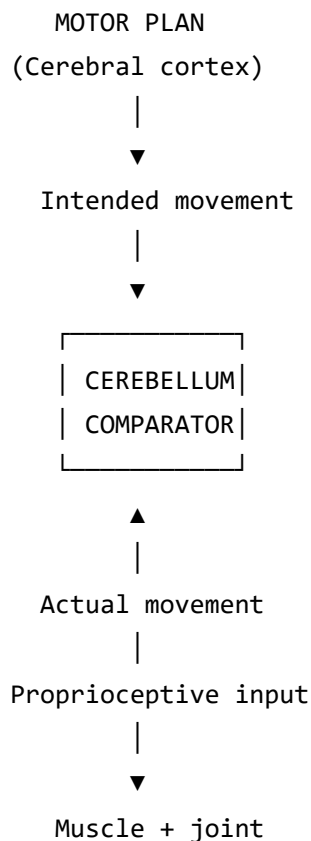
Sequential + Timed movements

B. Skilled Voluntary Movement

- Cerebellar hemisphere helps coordinate **skilled voluntary movements**.
- Particularly important for **timing and precision**.

4. CEREBELLUM AS A "COMPARATOR"

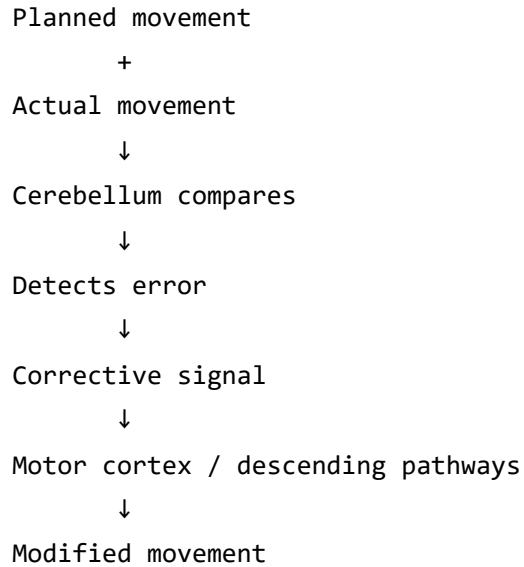
Basic Concept



Cerebellum receives

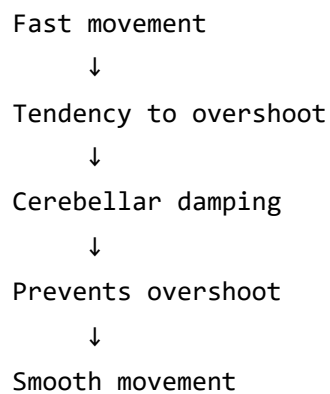
- Motor commands from **cerebral cortex**.
- Proprioceptive information through **spinocerebellar tracts**.
- Special sensory information:
 - Visual
 - Auditory
 - Vestibular

Then



Therefore: Cerebellum acts as a **comparator / servo mechanism**.

5. PREVENTS OVERSHOOT & TREMOR



- Prevents **overshoot**.
- Prevents excessive oscillation/tremor.
- Acts as a **damping system**.

6. MOTOR LEARNING

- Cerebellum compares:
 - **Ongoing movement**
 - **Required movement**
- Generates corrective signals → improves performance.

Long-term role

- Learning of **motor skills**.
- Long-term adjustment of motor performance.

Important pathway

Climbing fibres

↓

Purkinje cells

↓

Complex spikes

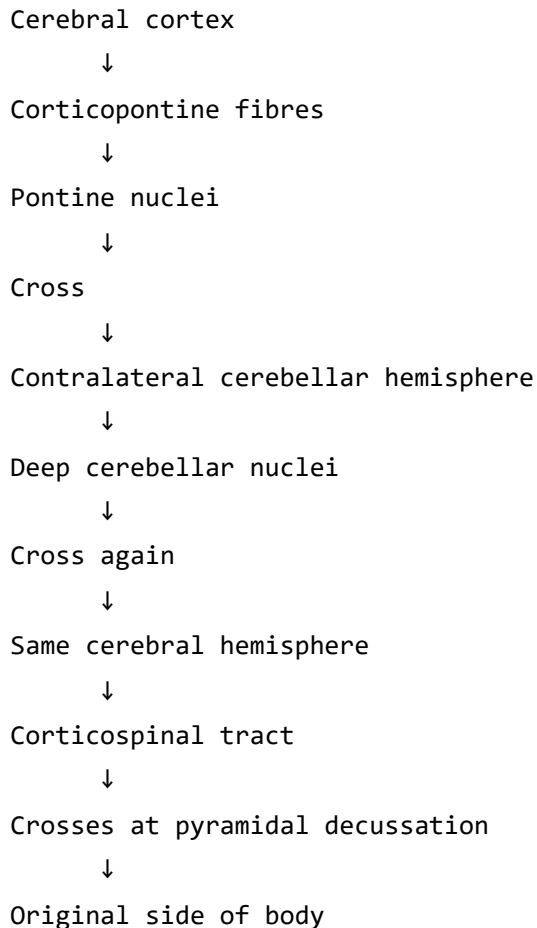
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Motor learning

- Climbing fibre activity increases during learning of a **new activity**.

7. CONTROL OF IPSILATERAL MOVEMENTS

Why does each cerebellar hemisphere control the same side?



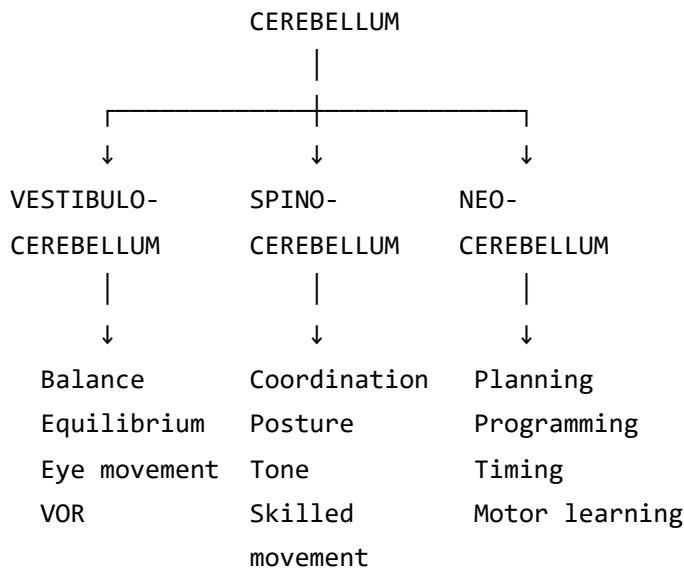
Result

Each cerebellar hemisphere controls coordination of movements on the ipsilateral side of the body.

Reason → DOUBLE DECUSSATION.

8. OVERALL FUNCTIONS — QUICK

REVISION



One-line Memory

Vestibulo → Balance + Eyes

Spino → Posture + Ongoing movement + Tone

Neo → Plan + Program + Precision + Learning

Other Overall Functions

- **Comparator / servo mechanism**
- **Smoothens movements**
- **Prevents overshoot and tremor**
- **Controls muscle tone**
- **Motor learning**
- **Coordinates movements of the ipsilateral side**